Replication of the wage equation in Lagomarsino, E., & Spiganti, A. (2020).

No gain in pain: psychological well-being, participation, and wages in the BHPS.

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Section 1. Data Preparation

Section 2. Estimation

Section 3. Analysis and Comparison

Section 4. Reflections

Section 5. Appendix 1: Tables of Results

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Data Preparation

The provided dataset consisted of 358,885 observations across waves 2 to 9 of the UK Household Longitudinal Study (UKHLS). Consistent with the approach of Lagomarsino and Spiganti (2020), the sample was restricted to individuals aged 16 to the UK state pension age, currently set at 66 for both genders. The analysis was further limited to individuals engaged in paid employment, excluding the self-employed and unemployed. This exclusion aimed to mitigate potential biases caused by differential survey participation rates.

Following the filtering, the sample was reduced to 166,314 observations. The majority (99.89%) were in paid employment, aligning with expectations given the initial inclusion criteria. Nevertheless, an issue emerged regarding missing data. Notably, 89.43% of responses for the variable 'year started current job' were missing. Since 99.89% of the sample was employed, a high response rate for the job start date was expected. This variable was intended to derive the 'Experience' variable, included within the wage equation of Lagomarsino and Spiganti (2020). Due to the extensive missing data, which compromised the reliability of this variable, I decided to omit 'Experience' and its polynomial terms from the analysis. Preserving the remaining sample size and ensuring data integrity took priority.

Variables from the original wage equation by Lagomarsino and Spiganti (2020), extracted from the BHPS, were recreated using corresponding variables available within the UKHLS dataset. Due to limitations in the available data, variables 'Union at workplace', 'member of union', 'Kids (0-2 years)', 'Kids (3-11 years)', and 'Teens (12-14 years)' could not be created. Instead, a 'number of children' variable was generated to approximate age-specific categories.

The dependent variable was specified as an inverse hyperbolic sine (IHS) transformation of average hourly wage. This transformation was chosen because it effectively addresses the skewness characteristic of wage distributions and accommodates values of zero, ensuring better statistical properties in the regression analysis.

Contrasting with the unbalanced panel structure of Lagomarsino and Spiganti (2020), a balanced panel was utilised. Only individuals observed in all eight waves were retained, which reduced the sample to 26,680 individuals, including 12,132 males and 14,548 females. Age was validated for logical progression across waves, and where discrepancies appeared, ages were corrected to increase incrementally by one year between waves. Gender was fixed for each individual across all waves, with any inconsistent entries rectified by assigning the initial wave gender to subsequent waves.

This process resulted in a balanced panel dataset suitable for replicating Lagomarsino and Spiganti's (2020) wage and mental health analysis, ensuring data comparability and analytical validity.

Estimation

The analysis uses panel data estimators: Pooled OLS (POLS), Random Effects (RE), Fixed Effects (FE), and Correlated Random Effects (CRE), based on an adaption of Mincer's wage equation included in Lagomarsino and Spiganti (2020):

$$IHS(wage) = \beta 0 + \beta 1 GHQ_{it} + \beta 2 X_{it} + \eta_t + \mu_i + \varepsilon_{it}$$

IHS(wage) denotes IHS transformation of hourly wage of individual i at time t. GHQ measures psychological well-being, X_{it} is a vector of explanatory variables, η_t accounts for time effects, μ_i captures individual heterogeneity, and ε_{it} is the idiosyncratic error.

POLS treats repeated observations of the same individual as independent, ignoring positive serial correlation caused by μ_i, therefore underestimating standard errors.

The RE model utilises a coefficient (1 – $\hat{\theta}$) for μ_i. $\hat{\theta}$ is random between 0 and 1, estimated using POLS residuals.

$$\theta = 1 - \sqrt{\frac{\sigma_\epsilon^2}{T\sigma_\mu^2 + \sigma_\epsilon^2}}$$

Consistent and efficient estimates of β and η_t are produced assuming covariance $(X_{it}, \mu_i) = 0$. However, this is unlikely to hold as μ_i will impact X_{it} as well as $IHS(wage)$.

FE removes μ_i through time-demeaning, therefore consistently estimates β. It focuses on within-individual variation and omits time-invariant variables.

CRE augments RE by including individual means of $\bar{X}_i$ and $\overline{GHQ}_i$ to model correlation between μ_i and regressors, consistently estimating β.

All models' standard errors are clustered by individual to account for serial correlation and heteroskedasticity.

The Regression-Based Hausman Test statistically determines whether FE or RE is appropriate. If μ_i is uncorrelated with regressors, RE is efficient. If correlated, FE is preferred.

Results, separate for men and women, are presented in Appendix 1.

Analysis and Comparison

Our main independent variable follows the IHS transformation proposed by Lagomarsino and Spiganti (2020) :

Log Mental Health$_{it}$ = $\ln(h_{it} + \sqrt{h_{it}^2 + 1})$

Where h_{it} denotes the inverted GHQ Caseness score, ranging from 0 (most distressed) to 12 (least distressed). This transformation reduces skewness and smooths the relationship with the dependent variable, while avoiding the issue of taking the logarithm of zero.

The estimated relationship between mental health and wages varies notably across panel data specifications. The POLS model shows a significant positive association for both males ($\beta 1$= 0.0555, p<0.01) and females ($\beta 1$ = 0.0256, p<0.01), implying that individuals with better mental health earn higher wages. However, POLS does not control for unobserved heterogeneity, so the estimates are biased if COV (X_{it}, μ_i) $\neq$ 0, where X_{it} includes GHQ_{it}.

The RE estimator yields smaller coefficients, 0.0132 (p<0.1) for males and 0.00793 (p<0.1) for females, indicating that POLS overestimates the effect. Nevertheless, if assumption COV (X_{it}, μ_i) = 0 does not to hold, the RE estimator remains inconsistent.

A Breusch-Pagan test was performed to determine whether the RE model is preferred over POLS.

H_0 : σ_μ^2 = 0.

The null hypothesis was rejected for both genders (p = 0.0000), suggesting the superiority of RE over POLS.

When unobserved heterogeneity is removed via fixed effects, the estimated coefficients decrease further and lose statistical significance (β_1 = 0.00907 for males, β_1 = 0.00395 for females). The FE estimator, which relies purely on within-individual variation, implies that once time-invariant characteristics are controlled for, changes in mental health have no significant effect on wages. The CRE model produces estimates of β that are identical to the FE model, due to modelling μ_i to be a function of the individual specific means of the time-varying regressors.

Although statistically insignificant, the FE coefficients suggests that a one-unit improvement in Log Mental Health$_{it}$ corresponds to approximately a 0.9% wage increase for males and a 0.4% increase for females, ceteris paribus.

The behaviour of the other coefficients further supports this interpretation. 'Almost' time-invariant variables such as education, occupation, and region exhibit strong effects in POLS and RE but become insignificant in FE, confirming they are biased by unobserved individual effects. In contrast, genuinely time-varying variables such as part-time work, firm size, and age retain significance, consistent with within-person variation. In FE White is omitted due to 0 within effect and wave is omitted due to perfect multicollinearity with age.

To formally test the validity of the RE model, we conducted a regression-based Hausman test, comparing the RE and FE models. After running the CRE model we can conduct a joint test of the individual-specific means.

H_0 : Time-varying regressors are uncorrelated with μ_i.

The results are highly significant for both males (test-statistic = 569.4, p = 0.0000) and females (test-statistic = 604.49, p = 0.0000), leading to a rejection of the null hypothesis. This provides strong evidence that the RE estimator is biased and that the FE or CRE models offer more reliable estimates.

Reflections

This replication mirrors Lagomarsino and Spiganti (2020) but reveals notable differences arising from sample composition, model specification, and the time period of analysis. While the original study employed POLS, FE, 2SLS, FE-2SLS, and SW estimators, this replication used POLS, RE, FE, and CRE. The most direct comparison is between the POLS and FE results, being methodologically comparable across studies.

In both the original BHPS analysis and this UKHLS replication, the Log Mental Health coefficient is positive across all specifications for both genders, indicating better psychological well-being correlates with higher wages. In the replication, this effect is statistically significant in the POLS model for both genders (β_1 = 0.0555, p<0.01 for males) (β_1 = 0.0256, p<0.01 for females), whereas Lagomarsino and Spiganti (2020) found only statistical significance for females (β_1 = 0.006, p>0.1 for males) (β_1 = 0.015, p<0.01 for females). Under the FE estimator, the coefficient is statistically insignificant for both genders in both studies, though remains positive. This consistency across datasets strongly suggests that once individual fixed effects are controlled for, the relationship between mental health and wages is largely driven by time-invariant personal attributes such as ability rather than changes in psychological well-being.

Key structural and methodological factors explain the differences in magnitude and precision. The replication omits key regressors from the original equation (experience polynomials, union variables, and age-specific child categories), potentially increasing omitted-variable bias in POLS. The replication sample size is smaller (12,132 males and 14,548 females) than the original analysis (26,194 males and 27,363 females) due to the balanced panel restriction, likely retaining more stable and healthier individuals, reducing mental health variability. The replication covers the period post 2008, marked by wage stagnation and employment volatility, potentially impacting the association between wage and well-being.

Overall, both studies converge on the conclusion that POLS overstate the true causal effect of mental health on wages. The decrease of significance under FE reinforces the importance of controlling for unobserved heterogeneity in psychological and labour market analysis.

Throughout the analysis, I utilised AI as a tutor, explaining concepts in detail and providing counterarguments for my ideas. If a clear command was given, the output was mostly useful, however fact checking AI was necessary. I found AI most useful as a proofreader, suggesting ways to rephrase my writing to remove unnecessary words or correcting grammar.

Appendix : Table of results

Table 1 : Variable definitions

Dependent variable	Label
IHS of Hourly Wage	ISH transformation of average hourly wage
Variables	Labels
Log Mental Health	Log transformation of the inverted Caseness score
Age	Age at date of interview in years
Age square	Square of age at date of interview /100
Age cube	Cube of age at date of interview /100
Has a degree	1 if 1st or higher degree
Children	Number of own children in the household
London	1 if living in London
White	1 if white
Widowed	1 if widowed or survive civil partner
Divorced or separated	1 if divorced, separated, or dissolved/separated from civil partner
Never Married	1 if never married
Private Sector	1 if employed in Private Sector
Professional	1 if professional
Manager	1 if managerial
Skilled non-manual	1 if skilled non-manual
Skilled Manual	1 if skilled manual
Part-Time Job	1 if part-time
Number of Employees	Number of employees at workplace (max 1,000) as average of the categories (e.g. “1-2” is rescored as 1.5)
Job Training (lag)	1 if received education or training related to current employment in the previous year

Table 2 : Wage equation males

	POLS	RE	FE	CRE
Log mental health	0.0555***	0.0132*	0.00907	0.00907
	(0.0118)	(0.00711)	(0.00705)	(0.00706)
age	0.125***	0.162***	0.215***	0.215***
	(0.0351)	(0.0254)	(0.0280)	(0.0281)
Age squared	-0.249***	-0.316***	-0.375***	-0.375***
	(0.0857)	(0.0619)	(0.0676)	(0.0676)
Age cubed	0.00166**	0.00201***	0.00238***	0.00238***
	(0.000680)	(0.000490)	(0.000530)	(0.000530)
Has a degree	0.188***	0.221***	-0.00278	-0.00278
	(0.0203)	(0.0223)	(0.0470)	(0.0471)
Children	0.0333***	0.0142**	0.00315	0.00315
	(0.00929)	(0.00580)	(0.00621)	(0.00622)
London	0.175***	0.0902**	-0.125**	-0.125**
	(0.0305)	(0.0381)	(0.0613)	(0.0614)
white	0.177***	0.147***	0	0.165***
	(0.0362)	(0.0436)	(.)	(0.0355)
widowed	-0.191***	-0.0166	0.0256	0.0255
	(0.0739)	(0.0227)	(0.0285)	(0.0285)
Divorced or separated	-0.0317	-0.0102	0.00975	0.00974
	(0.0354)	(0.0192)	(0.0204)	(0.0204)
Never married	-0.0728***	-0.0276**	0.00654	0.00654
	(0.0196)	(0.0133)	(0.0147)	(0.0148)
Private sector	0.0433**	0.0196	0.0171	0.0171
	(0.0173)	(0.0120)	(0.0140)	(0.0140)
Professional	0.414***	0.142***	-0.0164	-0.0164
	(0.0389)	(0.0300)	(0.0353)	(0.0354)
Manager	0.395***	0.133***	-0.00104	-0.00104
	(0.0264)	(0.0227)	(0.0259)	(0.0260)
Skilled non manual	0.190***	0.0456	-0.0247	-0.0247
	(0.0289)	(0.0279)	(0.0331)	(0.0331)
Skilled manual	0.131***	0.0183	-0.00946	-0.00946
	(0.0261)	(0.0191)	(0.0213)	(0.0213)
Part time job	-0.0271	0.236***	0.283***	0.283***
	(0.0650)	(0.0690)	(0.0744)	(0.0744)
Number of employees	0.000211***	0.000106***	0.0000381	0.0000381
	(0.0000220)	(0.0000211)	(0.0000272)	(0.0000272)
Job training lag	0.0289**	0.00276	-0.000553	-0.000553
	(0.0112)	(0.00533)	(0.00534)	(0.00535)
wave	0.0206***	0.0222***	0	
	(0.00171)	(0.00163)	(.)	

Table 2 continued

Mean log mental health				0.122***
				(0.0300)
Mean age				-0.0900*
				(0.0532)
Mean age squared				0.121
				(0.127)
Mean age cubed				-0.000627
				(0.000988)
Mean Has a degree				0.166***
				(0.0514)
Mean Children				0.0363***
				(0.0121)
Mean London				0.303***
				(0.0684)
Mean white				0
				(.)
Mean widowed				-0.351**
				(0.161)
Mean divorced or separated				-0.0354
				(0.0507)
Mean never married				-0.0773***
				(0.0270)
Mean Private sector				0.0306
				(0.0250)
Mean Professional				0.497***
				(0.0581)
Mean Manager				0.468***
				(0.0408)
Mean skilled non manual				0.259***
				(0.0463)
Mean skilled manual				0.173***
				(0.0392)
Mean part time job				-0.493***
				(0.110)
Mean Number of employees				0.000189***
				(0.0000374)
Mean job training lag				0.0940***
				(0.0332)
constant	0.307	0.0937	-0.907**	0.135
	(0.468)	(0.344)	(0.383)	(0.650)
R^2	0.321		0.129	
Observations	12132	12132	12132	12132

White and wave omitted in fixed effects because of collinearity.
Cluster robust standard errors reported in parentheses.* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 : Wage equation females

	POLS	RE	FE	CRE
Log mental health	0.0256***	0.00793*	0.00395	0.00396
	(0.00734)	(0.00469)	(0.00483)	(0.00483)
age	0.0587**	0.0925***	0.121***	0.121***
	(0.0269)	(0.0245)	(0.0283)	(0.0283)
Age squared	-0.0950	-0.182***	-0.194***	-0.194***
	(0.0650)	(0.0598)	(0.0681)	(0.0681)
Age cubed	0.000446	0.00114**	0.00123**	0.00123**
	(0.000506)	(0.000471)	(0.000533)	(0.000533)
Has a degree	0.177***	0.222***	0.0671**	0.0671**
	(0.0167)	(0.0157)	(0.0281)	(0.0281)
Children	0.00358	-0.0116*	-0.0147*	-0.0147*
	(0.00885)	(0.00668)	(0.00782)	(0.00783)
London	0.190***	0.188***	0.123**	0.123**
	(0.0293)	(0.0270)	(0.0530)	(0.0531)
white	0.0678**	0.0548*	0	0.0691**
	(0.0275)	(0.0302)	(.)	(0.0272)
widowed	0.0129	-0.0165	-0.0346	-0.0345
	(0.0435)	(0.0352)	(0.0387)	(0.0387)
Divorced separated	-0.0206	-0.00907	0.00290	0.00298
	(0.0185)	(0.0128)	(0.0157)	(0.0157)
Never married	-0.0149	-0.00910	-0.00834	-0.00831
	(0.0164)	(0.0119)	(0.0148)	(0.0148)
Private sector	-0.0336**	-0.0291***	-0.0151	-0.0151
	(0.0144)	(0.0108)	(0.0132)	(0.0132)
Professional	0.602***	0.333***	0.130***	0.130***
	(0.0371)	(0.0349)	(0.0451)	(0.0452)
Manager	0.393***	0.238***	0.113***	0.113***
	(0.0208)	(0.0193)	(0.0238)	(0.0238)
Skilled non manual	0.165***	0.0924***	0.0234	0.0234
	(0.0177)	(0.0181)	(0.0237)	(0.0237)
Skilled manual	0.0293	0.0285	0.00828	0.00828
	(0.0237)	(0.0221)	(0.0247)	(0.0247)
Part time job	-0.0146	0.105***	0.150***	0.150***
	(0.0142)	(0.0118)	(0.0137)	(0.0137)
Number of employees	0.000195***	0.000147***	0.0000788***	0.0000788***
	(0.0000188)	(0.0000171)	(0.0000237)	(0.0000237)
Job training lag	0.00111	0.00435	0.00261	0.00262
	(0.00899)	(0.00508)	(0.00516)	(0.00516)
wave	0.0222***	0.0231***	0	
	(0.00142)	(0.00139)	(.)	

Table 3 continued

Mean log mental health				0.0502***
				(0.0175)
Mean age				-0.103**
				(0.0446)
Mean age squared				0.197*
				(0.106)
Mean age cubed				-0.00154*
				(0.000810)
Mean Has a degree				0.0879***
				(0.0335)
Mean Children				0.0317**
				(0.0133)
Mean London				0.0615
				(0.0615)
Mean white				0
				(.)
Mean widowed				0.0615
				(0.0622)
Mean divorced or separated				-0.0248
				(0.0272)
Mean never married				-0.00738
				(0.0238)
Mean Private sector				-0.0197
				(0.0218)
Mean Professional				0.538***
				(0.0639)
Mean Manager				0.311***
				(0.0333)
Mean skilled non manual				0.150***
				(0.0311)
Mean skilled manual				0.00671
				(0.0386)
Mean part time job				-0.234***
				(0.0250)
Mean Number of employees				0.000120***
				(0.0000318)
Mean job training lag				-0.0149
				(0.0253)
Constant	1.358***	1.087***	0.259	1.899***
	(0.358)	(0.327)	(0.385)	(0.484)
R^2	0.359		0.096	
Observations	14548	14548	14548	14548

White and wave omitted in fixed effects due to collinearity.
Cluster robust standard errors reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 : Male regression based Hausman test

Correlated random effects model with standard errors clustered by individual.
Mean of white omitted because of collinearity.

Joint test of the individual specific means.

Test Statistic	degrees of freedom	p value
569.41	18	0.0000

H_0 : Time-varying regressors are uncorrelated with μ_i.

H_1 : Time-varying regressors are correlated with μ_i.

P < 0.01 provides sufficient evidence to reject the null hypothesis and suggest that fixed effects are preferred over random effects.

Table 5 : Female regression based Hausman test

Correlated random effects model with standard errors clustered by individual.
Mean of white omitted because of collinearity.

Joint test of the individual specific means.

Test Statistic	degrees of freedom	p value
604.49	18	0.0000

H_0 : Time-varying regressors are uncorrelated with μ_i.

H_1 : Time-varying regressors are correlated with μ_i.

P < 0.01 provides sufficient evidence to reject the null hypothesis and suggest that fixed effects are preferred over random effects.